

Utkarsh Tripathi

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Education

San Jose State University, USA | Master of Science in Data Analytics

Aug 2024 - Present

Bharati Vidyapeeth Deemed (to be) University, India | Bachelor of Electronics (3.57)

Jul 2014 - Jun 2018

Work Experience

Retired.com — WAO Management LLC

Jun 2025 – Dec 2025

Data Engineer Intern & Co-Op | Project – Domo Dash Tech.

- Built and deployed end-to-end pipelines from **ADO APIs/queries** into **Snowflake** using **Rivery**, **dbt**, **stored procedures**, powering **Domo dashboards** for sprint planning and KPI tracking and enabling downstream analytics/AI use cases; used **Postman** for API testing and **eliminated 98% of manual Excel work**, improving capacity visibility and sprint predictability.
- Optimized **Rivery job efficiency** (1 row/credit → 200 rows/credit), integrated **HubSpot CRM** pipelines for IT migration, and implemented **Azure CI/CD** with automated testing, **cutting deployment time by 90%** and reducing compute costs while keeping data continuously available for reporting and model development.
- Debugged complex live production **Snowflake/Rivery job failures** within SLAs and prototyped a hackathon **MVP AI agent** that analyzes pipeline logs, recommends retries, and flags likely root causes, contributing to a **40% reduction in downtime** on critical data jobs.

Cognizant Technology Solutions Pvt. Ltd, India

Apr 2021 - Aug 2024

Data Engineer (Associate) | Sanofi Pharma Aventis

- Designed and **automated BI dashboards** using **Power BI**, **QlikSense**, and **SQL** by integrating data from **CRM**, **ERP**, and **warehouse systems**, helping optimize business intelligence reporting applications.
- Built **data migration** projects to **Snowflake** and **AWS S3**, **optimizing ETL pipelines**, **reducing query response times by 40%**, and improving data accessibility for large-scale analytics.
- Led a team of three interns, **improving data pipelines** and optimizing system performance of applications in **IICS**, **Power BI** and **Snowflake**, **resulting in a 75% efficiency gain**.

Cognizant Technology Solutions Pvt. Ltd, India

Nov 2018 - Mar 2021

Business Intelligence Engineer (Programmer Analyst / Trainee) | Sanofi Pharma Aventis

- **Migrated BI infrastructure** from **QlikSense** to **Power BI** with **Amazon S3** and **Snowflake** data warehouse solutions built in **Informatica**, designing optimized schemas and pipelines that **improved data availability by 90%** and delivered **85% faster query performance** for global pharmaceutical sales and vaccine analytics dashboards.
 - **Automated ETL workflows** and health-monitoring scripts in **Python** and **SQL** for 20+ sales, marketing, and revenue management applications, eliminating manual data processing, accelerating ingestion, and **maintaining 99.5% uptime**.
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Academic Projects

Multi-Agent | AI-Powered Autonomous Workforce Assistants ([REPO Link](#))

Aug 2025 – Present

Currently building a multi-agent “chain-of-debate” LLM assistant over 106k Slack messages, 50k cleaned Enron emails and 50k-token AMI transcripts, orchestrated with **Apache Airflow** → **AWS S3** → **Snowflake**; the prototype produces structured **JSON/ICS/JSONL** outputs with **97.82% JSON validity**, **94.09% F1** for attendee extraction, **ROUGE-1 = 0.85**, and a **9.87% improvement in overall quality** over a single-agent baseline.

Generative AI | Emotion-Aware Transformer Text-to-Speech

Jan 2025 – May 2025

Implemented from scratch an **emotion-conditioned Transformer TTS** model in **PyTorch** that maps text and discrete emotion labels to mel-spectrograms and reconstructs audio with a custom **Griffin-Lim** vocoder; built an evaluation harness using **Whisper ASR** for WER/CER, **ECAPA-TDNN** speaker embeddings, and **librosa** F0/RMS analysis, where neutral speech achieves **22.73% WER**, **10.37% CER** and distinct emotion-specific F0 ranges (~340–450 Hz).

Skills

Data Engineering & Cloud: Snowflake, dbt, Rivery, CI/CD, Azure Pipelines, Stored Procedures, AWS (S3, RDS, Elastic Beanstalk), Apache Spark, Hadoop, Kafka; **Programming & Databases:** Python, SQL, Shell Scripting, PostgreSQL, MySQL, Redshift, MongoDB, Oracle; **ML & Analytics:** PyTorch, scikit-learn, Snowflake ML, Spark ML, XGBoost, SVM, Logistic Regression; **Visualization & Tools:** Power BI, Tableau, QlikSense, Domo, Git, Postman, React, CSS, JS.

Certifications and Trainings

AWS Certified Cloud Practitioner (In Progress) | Training in Digital Marketing.